Putting It Together

Review and capstone: Messy Questions, Answered Out Loud

Stat 220 | Statistics for Data Science

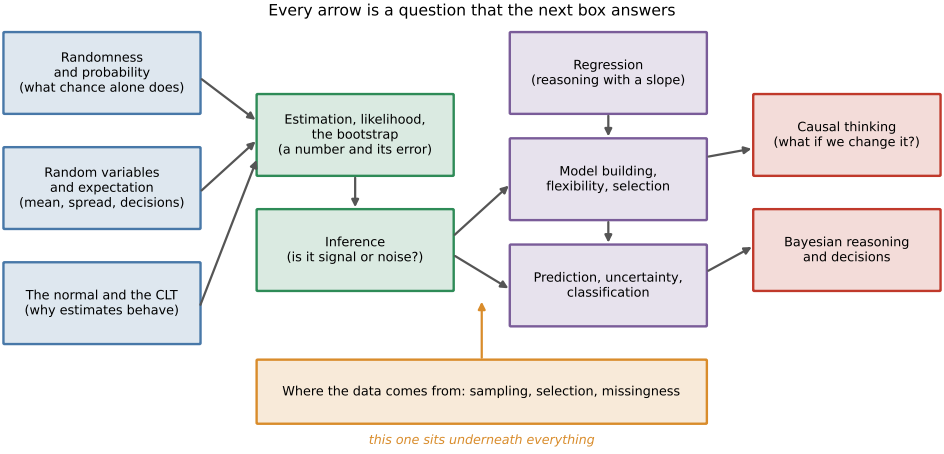
Where this fits

- You now have the tools. This part is about the skill that actually gets tested: taking an ambiguous, badly posed question and reasoning through it in front of someone.
- Nobody will ask you to derive a t-statistic. They will hand you a situation and watch how you think.
- The good news: the same handful of moves works on almost every question, and you have been practicing them all semester.

Principle: what is actually being evaluated

Not whether you know the formula. Whether you notice what is missing, say what you assume, and know what would change your answer.

The whole course on one page



Live experiment: the interview bracket

Live demo: two minutes, out loud, no notes

Pair up. Each pair draws two scenario cards.

1. Partner A reads a card. Partner B answers **out loud for two minutes**. No writing.
2. Partner A scores using the five-beat rubric on the next slide, one point per beat.
3. Swap, then the highest scorer from each pair advances.

Semifinalists answer at the front, and the class votes.

Materials: printed scenario cards (`Unit_12_Scenario_Cards.pdf`). Time: 30 minutes for the full bracket. The scoring rubric is deliberately not about being right. It is about the shape of the answer.

The five-beat answer

Use this structure on any question you are handed:

1. **Clarify.** What decision is this for, and what would count as an answer?
2. **State assumptions.** What do you need to be true, and which are you unsure of?
3. **Name the approach.** What you would compute, and why that and not something else.
4. **Name the traps.** What would make this answer wrong.
5. **Say what would change your mind.** What evidence would flip the recommendation.

Principle: beats 4 and 5 are where candidates separate

Almost everyone can do the middle. Very few volunteer how their own answer could be wrong, and that is precisely the mark of someone you can trust with a decision.

What this unit is

- No new mathematics. Everything here is a tool you already have. The work is choosing which one, fast, on a problem stated badly by someone who is not a statistician.
- Day 1 builds the reusable answer structure and the checklist. Day 2 runs six full scenarios. Day 3 is the bracket, where you do it out loud.
- The scenarios come from every unit at once, and none of them announce which unit they came from. That is the point.

The universal red-flag checklist

Run this on *any* number, chart, or claim someone hands you:

1. **Where did this data come from**, and who is missing from it?
2. **What is the denominator**, and is it comparable across the groups?
3. **How big is the sample**, and how many of the rare thing did we actually see?
4. **Where is the uncertainty**? Is there an interval, and is it honest?
5. **Was this the only comparison made**, or the winner of many?
6. **Is this association being read as a cause**, and what is the comparison group?
7. **What would this look like if nothing were happening?**

The end-to-end workflow, consolidated

1. **Pin the question** to a decision someone will make.
2. **Interrogate the data's origin** before you open it.
3. **Look at the data** (distributions, missingness, outliers) before modeling.
4. **Choose the simplest method the question needs**, and say why.
5. **Estimate, with uncertainty**, using a bootstrap when formulas run out.
6. **Check the assumption most likely to break**, and check it out of sample.
7. **Translate into a recommendation**, with the risk and one caveat.

Every module contributed one line to this list.

Scenario 1: the metric that jumped

They say: "Signups are up 22% this week. The new homepage is working. Roll it out everywhere."

Your turn: two minutes, out loud

What do you ask, what do you compute, and what would make you say yes?

Scenario 1: a strong answer

- **Clarify:** 22% of what base, over what window, and compared to which period?
- **Assumptions:** that nothing else changed that week. Check holidays, a marketing push, a tracking change, and last year's same week.
- **Approach:** if the homepage was launched to everyone at once, there is no control group and no causal estimate. Look for a staged rollout or a holdout region for a difference-in-differences comparison.
- **Traps:** week-to-week noise on a small base, a definition change, regression to the mean after a bad week, and the seasonality that affects everyone.
- **Changes my mind:** a randomized holdout showing the same lift.

Scenario 2: the impressive model

They say: "Our new churn model is 96% accurate and its AUC is 0.99. Can we start acting on it Monday?"

Your turn: two minutes, out loud

What is your first reaction, and what three things do you check before Monday?

Scenario 2: a strong answer

- **First reaction:** suspicion, not celebration. If churn is 4%, predicting "nobody churns" scores 96%, and an AUC of 0.99 on a human behavior problem usually means leakage.
- **Check 1:** the base rate and the trivial baseline. **Check 2:** whether any feature is knowable only *after* the outcome. **Check 3:** precision and recall at the threshold we will actually use.
- **Then the real question:** what action follows a flag, what does a false alarm cost, and what does a miss cost? That gives the threshold.
- **Changes my mind:** clean performance on a time-split holdout with no post-outcome features.

Scenario 3: the subgroup discovery

They say: "The campaign didn't work overall, but it was hugely effective for women aged 35 to 44 in the Midwest. Let's target them."

Your turn: two minutes, out loud

Is this a finding? What would you need to believe it?

Scenario 3: a strong answer

- This is a **multiple comparisons** problem. With a dozen demographic splits there are hundreds of subgroups, and the best one always looks impressive.
- Ask: was this subgroup specified *before* the analysis? How many were examined? What is the sample size inside it?
- Expect **regression to the mean**: the effect will shrink substantially on new data, even if something real is there.
- **What would convince me:** the same subgroup effect replicating in a fresh campaign, preregistered.

In practice: the line to remember

"Subgroup findings are hypotheses, not conclusions. I would treat this as a candidate to test, not a result to act on."

Scenario 4: the executive summary

They say: "Customers who use our mobile app spend 3 times more. We should push everyone to install the app."

Your turn: two minutes, out loud

What is wrong, and what would you propose instead?

Scenario 4: a strong answer

- **Selection, not effect.** People who install the app were already your most engaged customers. The app is a marker of loyalty at least as much as a cause of it.
- **Reverse causation is live too:** heavy spenders install the app *because* they shop often.
- **Proposal:** randomize an install prompt across a holdout, or use a staged rollout by region, and measure the change in spending rather than its level.
- **The honest interim answer:** adjusting for prior spending and tenure shrinks the 3-times gap substantially, and whatever remains is still not an effect.

Scenario 5: the survey

They say: "Our NPS survey came back at 42, up from 38. Customer satisfaction is improving."

Your turn: two minutes, out loud

What do you need to know before you believe the four-point move?

Scenario 5: a strong answer

- **How many responded, and out of how many asked?** A 4-point move on 200 responses is well inside noise.
- **Did the responding population change?** If the survey went out after a service outage last quarter and not this one, you are comparing different people.
- **Did the instrument change?** Question wording, placement, and timing all move scores without moving satisfaction.
- **What I would do:** attach an interval to both numbers, compare responders to non-responders on the variables we do have, and corroborate with behavior, such as renewal and usage.

Scenario 6: the forecast

They say: "The model predicts $4.2M in revenue next quarter. Put it in the board deck."

Your turn: two minutes, out loud

What do you add before that number goes in front of the board?

Scenario 6: a strong answer

- **An interval, sized from real out-of-sample errors**, not from the training fit. A point forecast with no range is half an answer.
- **Which layer of error dominates:** noise, estimation, model structure, or shift? For a quarterly forecast in a changing market, shift usually dominates, and no in-sample statistic warns you.
- **Whether we are extrapolating:** is next quarter inside the range of conditions the model was fit on?
- **One sentence of caveat:** the conditions under which this number should not be trusted.

Principle: what to hand a board

The point, the range, the dominant source of error, and the one thing that would make it wrong.

Scenario 7: the model that decides about people

They say: "Our hiring model is 88% accurate at predicting who will be a top performer. Let's use it to screen applications."

Your turn: two minutes, out loud

What do you ask, what could go wrong, and what would you need before this touches a real applicant?

Scenario 7: a strong answer

- **The label:** "top performer" is a manager's rating, a proxy carrying whatever biases that manager had. The model predicts the rating, not the performance.
- **The sample:** it only observes people who were *hired*. Strong candidates the old process rejected are invisible, so the model learns the old process.
- **The metric:** 88% accuracy means nothing without the base rate. If 85% of hires are rated acceptable, the model is barely beating "say yes."
- **Before deployment:** the base rate, precision and recall at the operating threshold, error rates broken out across groups, and a human in the loop.

Scenario 8: the number with no error bar

They say: "Customer lifetime value is $1,840. I've put it in the pricing model."

Your turn: two minutes, out loud

Where did that number come from, and what are the three things you check before pricing depends on it?

Scenario 8: a strong answer

- **Mean or median?** Lifetime value is heavily right-skewed. If $1,840 is a mean, a handful of whales are setting your prices for everyone.
- **Which customers?** Almost certainly the ones who have already churned, since only they have a complete lifetime. That is survivorship in reverse: your best customers are excluded precisely because they are still here.
- **How wide is the interval?** A bootstrap on a skewed quantity like this is often embarrassingly wide, and pricing built on a point estimate inherits all of that.
- **What I would report:** median and mean, an interval on each, and a segment breakdown, because one number for all customers is the actual error here.

Translating for a non-technical audience

- **Lead with the decision**, not the method. "Ship it" or "do not ship yet, and here is what would settle it."
- **Quantify the risk in their units:** dollars, customers, hours, not p-values.
- **Use natural frequencies:** "about 6 of every 100" beats "a 6% rate," and both beat "an 18% relative increase."
- **Give one caveat, not five.** The most likely way to be wrong, in one sentence.

Trap: two opposite failures

Drowning the room in caveats is as useless as hiding all of them. Pick the one that could change the decision.

When you do not know

- Say so, then keep going: "I do not know, but here is how I would find out." That answer scores well, and pretending scores zero.
- Reason out loud. The interviewer is watching the process, not waiting for a number.
- Anchor on what you do know: the base rate, the sample size, the comparison group, what chance alone would produce.

Principle: the honest default

"It depends" is a fine start, as long as the next sentence says *on what*, and the one after that says what you would check first.

LLM check: mistake or not? (1 of 2)

LLM check: we gave an LLM a full interview question

We told it: "Our conversion rate dropped from 4.1% to 3.8% last month. What happened?"

It answered: "A 0.3-point drop is a 7.3% decline in conversion. Likely causes include increased competition, seasonal effects, and site performance issues. I would recommend auditing your checkout flow and running a competitor analysis."

Mistake, or no mistake?

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Mistake, or no mistake?

Trap: verdict: mistake

It jumped straight to explanations without asking whether there is anything to explain. No sample size, no interval, no check on whether tracking or the definition changed. It is generating plausible stories for what may be noise.

LLM check: mistake or not? (2 of 2)

LLM check: same question, a different answer

We told it: "Our conversion rate dropped from 4.1% to 3.8% last month. What happened?"

It answered: "First, how many visitors? At 5,000 a month that difference is well within noise, and at 5 million it is not. Before looking for causes I would check whether tracking or the conversion definition changed, whether traffic mix shifted, and how this month compares to the same month last year. If it survives all that, then we look for causes."

Mistake, or no mistake?

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Mistake, or no mistake?

Principle: verdict: no mistake

Sample size first, instrument second, mix third, causes last. That ordering is the entire course in four sentences.

The whole workflow

1. **Pin the question** to a decision somebody will make.
2. **Interrogate the data's origin** before you open it: population, frame, who is missing.
3. **Look at the data**: distributions, missingness, outliers, and what one observation is.
4. **Choose the simplest method the question needs**, and say why that one.
5. **Estimate with uncertainty**, using a bootstrap when no formula applies.
6. **Check the assumption most likely to break**, out of sample.
7. **Translate into a recommendation** with the risk and one caveat attached.

The master trap list (1 of 2)

Trap: the ones that come up most

- Ignoring the base rate after a positive test or a model flag.
- Reporting accuracy on imbalanced data, or a rate with no denominator.
- Quoting a relative change with no absolute change.
- Confusing $P(A \mid B)$ with $P(B \mid A)$.
- Reporting a point estimate with no interval, or an interval that is too narrow.
- Treating the winner of many comparisons as a discovery.
- Counting rows instead of independent units.

The master trap list (2 of 2)

Trap: the ones that end careers

- Reading a regression coefficient as the effect of an intervention.
- Controlling for a collider or a mediator.
- Judging a model on training data, or tuning on the test set.
- Believing a near-perfect model instead of hunting for leakage.
- Treating a big sample as a representative one.
- Dropping missing rows without asking why they are missing.
- Extrapolating past the data, and assuming the future matches the past.

The final question bank (1 of 2)

Answer each in two minutes, out loud, using the five beats.

1. A test is 99% accurate and you tested positive for a rare disease. Now what?
2. Our A/B test shows a 40% lift with $p = 0.04$ on 500 users. Do we ship?
3. Why is a prediction interval wider than a confidence interval?
4. Users of feature X retain better. Should we push everyone onto it?
5. Our model is 96% accurate. Are you impressed?
6. How would you put error bars on a forecast?
7. Should you control for everything you have? Explain.
8. Our data is skewed. Can we still use a t-test?

The final question bank (2 of 2)

9. We have 12 million rows. Do we still need to think about sampling?
10. How would you set the threshold on a fraud model?
11. Our best rep slumped last quarter. Did the new comp plan fail?
12. A subgroup shows a huge effect. Do we target it?
13. Explain the bootstrap in thirty seconds.
14. What would you check before explaining a spike in a dashboard?
15. How do you know you are not overfitting?
16. What is the single most important question to ask about any dataset?

The mastery checklist for the whole course

You are ready when you can, without notes:

- say what chance alone would produce, before explaining any pattern
- attach an honest interval to any estimate, formula or not
- translate any accuracy claim into the conditional probability someone needs
- tell an association from a causal claim, and name what design would earn the second
- judge a model on data it has never seen, and say what would break it
- trace any dataset back to the process that made it, and name who is missing
- turn all of that into a recommendation with a stated risk and one caveat

Running the bracket (logistics)

- **Round 1 (pairs, 10 min).** Everyone answers one card, partner scores the five beats. Swap. Higher score advances.
- **Round 2 (fours, 10 min).** Winners pair off, and the two losers score.
- **Semifinals and final (front of room).** Two minutes each, class votes with the rubric, not on charisma.
- **The rule that matters:** nobody may say "run a t-test" without saying what would make the test inappropriate.

Principle: what wins

Not the most confident answer. The one that names its own assumptions and says what would change its mind.

What you should be able to do with software

Nobody is asking you to memorize syntax. You are expected to know *which* procedure the question calls for, what to hand it, and what the output means. With an AI assistant open, you should be able to produce any of these and defend the result.

You should be able to	**What you would reach for**
Everything from Units 1 through 10, on demand	you should not need to be told which
Load, inspect, and audit a new dataset	`read_csv`, `describe`, `isna`
Compare two groups honestly	`ttest_ind`, plus an interval
Fit and validate a model	`cross_val_score` inside a pipeline
Attach an interval to any estimate	bootstrap, or held-out residuals
Turn a prediction into a decision	expected cost over the predictive distribution
Say what the analysis cannot support	no library for this one

If you cannot say what the output means, the fact that you produced it is worth nothing.

The last slide

- Computation is cheap and getting cheaper. A model can be fit by anyone, in one line, in seconds.
- What does not get cheaper is knowing which question to ask, which number is meaningless, and when the confident answer on the screen is wrong.
- That judgment is the job. It is what you have been building all semester, and it is what every one of those interview questions is really testing.

Principle: the whole course, in one sentence

Anyone can produce a number. You have been trained to ask what would have to be true for that number to mean anything.